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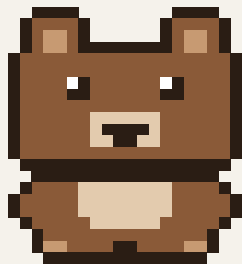
Policy Evaluation Lab

Lecture 2: Causality (II)

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PART 1

Khuzdar and Maria



Notation

In this lecture we use the notation of Angrist and Pischke:

- $Y_{1i} \equiv Y_i(1)$: potential outcome of individual i if treated ($D_i = 1$)
- $Y_{0i} \equiv Y_i(0)$: potential outcome of individual i if not treated ($D_i = 0$)
- Y_i : actual (observed) outcome



Example: Khuzdar and Maria

Two visiting students at MIT have to decide whether to opt in to the university health insurance (HI) plan by paying a fee.

- **Khuzdar** Khalat, from Kazakhstan, has national HI that covers all its citizens. His health is weak to begin with, and he decides to buy the insurance
- **Maria** Moreno, from Chile, is very healthy and little concerned by Boston winters. She does not buy the insurance
- For each individual there are two potential outcomes, whose difference is the causal effect of HI:
 - for Khuzdar: $\tau_K = Y_{1K} - Y_{0K}$
 - for Maria: $\tau_M = Y_{1M} - Y_{0M}$



Example: Khuzdar and Maria

	Khuzdar	Maria
Potential outcome without insurance: Y_{0i}	3	5
Potential outcome with insurance: Y_{1i}	4	5
Treatment (insurance status chosen): D_i	1	0
Actual health outcome: Y_i	4	5
Treatment effect: $Y_{1i} - Y_{0i}$	1	0

Note: $i = K$ or M ; $D_i = 1$ or 0



Example: Khuzdar and Maria

- Only one of the two potential outcomes is revealed
- Khuzdar and Maria make different HI choices, and their actual outcomes Y_i differ:
 - Khuzdar's actual (realised and observed) health outcome is $Y_K = 4$
 - Maria's is $Y_M = 5$
- If we make a **naïve** comparison of the insured Khuzdar with the uninsured Maria, we get $Y_K - Y_M = -1$
- From this raw comparison between healthy Maria and frail Khuzdar, we could conclude that getting insurance is counterproductive



Example: Khuzdar and Maria – selection bias

The comparison of actual outcomes does not identify a causal effect. Linking observed and potential outcomes, and adding and subtracting $Y_{0K} = 3$:

$$Y_K - Y_M = Y_{1K} - Y_{0M} = \underbrace{(Y_{1K} - Y_{0K})}_{4-3} + \underbrace{(Y_{0K} - Y_{0M})}_{3-5} = -1$$

We get two hidden comparisons:

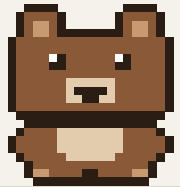
- $Y_{1K} - Y_{0K} = 1$: the causal effect of HI on Khuzdar, i.e. τ_K
- $Y_{0K} - Y_{0M} = -2$: the difference in health between Khuzdar and Maria if neither had HI

The second term shows a lack of comparability between K and M:
selection bias



PART 2

Health insurance and health



Case: effect of health insurance on health

- **Data: NHIS** (National Health Interview Survey)
 - annual survey of the US population, with specific information on health and HI
 - year used in the analysis: 2009; sample: married individuals aged 26–59
- **Outcome** Y_i : health index (1 poor, 2 fair, 3 good, 4 very good, 5 excellent)
- **Treatment** D_i : coverage by health insurance
 - treatment group: insured, $D_i = 1$
 - control group: uninsured, $D_i = 0$



Table 1: outcome of insured and uninsured couples

	Husbands			Wives		
	Some HI	No HI	Difference	Some HI	No HI	Difference
	(1)	(2)	(3)	(4)	(5)	(6)
A. Health						
Health index	4.01	3.70	.31	4.02	3.62	.39
	[.93]	[1.01]	(.03)	[.92]	[1.01]	(.04)

- A simple comparison of averages suggests that those with HI are healthier:
 - the index is higher by 0.31 for insured men and by 0.39 for insured women

Source: Angrist and Pischke (2015), *Mastering 'Metrics*, Table 1.1, panel A.



Causal effect?

- Is the health of an individual with HI any better than the health of the **same** person had they had no HI?
- Looking at the data, it seems so
- However, the result is misleading: we are not comparing apples with apples, but apples with oranges

Remember: a good **control group** should reveal the fate of the treated in the counterfactual world of no treatment



Balance check

- We need to analyse the two groups more deeply, with the help of statistical tools
- perform a **balance check**:
 - verify whether there are statistically significant differences in observable characteristics between the treated and the control group
- A **balanced** sample implies that the characteristics of the individuals in the treated and in the control group have the same mean



Test of statistical significance: t-test

- Using the sample averages, we compute the **t - statistic**: the difference in means divided by the estimated standard error of this difference

$$t_{stat} = \frac{\bar{X}^1 - \bar{X}^0 - 0}{\widehat{s.e.}(\bar{X}^1 - \bar{X}^0)}$$

- t_{stat} is compared with the critical values of Student's t distribution
- We apply the same test to check whether a coefficient from an estimation is statistically significant, i.e. whether we can reject that the effect is zero:

$$t_{stat} = \frac{coeff - 0}{s.e.(coeff)}$$

compared with the same critical values



T-tests: critical values

- if $|t_{stat}| > 2.58$, the difference is **statistically significant at 1%**, usually indicated with ***
- if $|t_{stat}| > 1.96$, the difference is **statistically significant at 5%**, usually indicated with **
- if $|t_{stat}| > 1.64$, the difference is **statistically significant at 10%**, usually indicated with *

If the **p-value** is provided, compare it with the significance levels: e.g. p-value < 0.01 means significance at 1%.

Significant at 1%: if the true difference were zero, a difference this large would arise by chance less than 1% of the time



Table 2: characteristics of insured and uninsured couples

	Husbands			Wives		
	Some HI (1)	No HI (2)	Difference (3)	Some HI (4)	No HI (5)	Difference (6)
<i>A. Health</i>						
Health index	4.01 [.93]	3.70 [1.01]	.31 (.03)	4.02 [.92]	3.62 [1.01]	.39 (.04)
<i>B. Characteristics (standard errors of the differences in parentheses)</i>						
Nonwhite	.16	.17	-.01 (.01)	.15	.17	-.02 (.01)
Age	43.98	41.26	2.71 (.29)	42.24	39.62	2.62 (.30)
Education	14.31	11.56	2.74 (.10)	14.44	11.80	2.64 (.11)
Family size	3.50	3.98	-.47 (.05)	3.49	3.93	-.43 (.05)
Employed	.92	.85	.07 (.01)	.77	.56	.21 (.02)
Family income	106,467	45,656	60,810 (1,355)	106,212	46,385	59,828 (1,406)
Sample size	8,114	1,281		8,264	1,131	

Source: Angrist and Pischke (2015), *Mastering 'Metrics*, Table 1.1.



Balance check

- In Table 2, brackets report the **standard deviations** (s.d.) of the sample means (cols. 1–2 and 4–5): a measure of variability in the data
- Parentheses report the **standard errors** (s.e.) of the differences (cols. 3 and 6): a measure of statistical precision

Test for age:

- $Age^T - Age^C = 43.98 - 41.26 \approx 2.71$
- $t_{stat} = \frac{2.71 - 0}{0.29} = 9.34 > 2.58$ (critical value at 1%)
- The difference is statistically significant at 1%

People in T and C do **not** have the same average age



Insured vs uninsured: characteristics

- Panel B of Table 2 shows significant differences between insured and uninsured (cols. 3 and 6): the insured are on average older, more educated and richer
- Most variables are highly correlated with both health (Y) and health insurance (D)
- More educated people are both healthier and more likely to be insured
- Therefore, the difference in health between insured and uninsured (0.31, significant at 1%) partly reflects the difference in education between the two groups

The uninsured are likely **not** a good control group for the insured: can we estimate a causal effect?



Causal effect?

- In panel A of Table 2, we see the average Y_{1i} only for the insured and the average Y_{0i} only for the uninsured
- This corresponds to the **difference in means** by treatment status
- Even in a hypothetical situation where the causal effect of insurance is zero ($\tau = 0$), we would find that insured individuals are healthier than uninsured ones, because they are on average more educated, richer, more likely to be employed...
- This leads to a **biased** estimate of the effect of HI on health

Difference in group means = Average causal effect + Selection bias



PART 3

Selection bias



Average causal effect (I)

The investigation of an average causal effect begins with the comparison of the average health of insured and uninsured individuals.

We can only observe the actual outcome in the two subsamples of size n_1 and n_0 :

- $Avg_n[Y_i \mid D_i = 1] = \frac{1}{n_1} \sum_{i=1}^{n_1} Y_{1i}$ among the insured, which is necessarily an average of Y_{1i} , with no information on Y_{0i}
- $Avg_n[Y_i \mid D_i = 0] = \frac{1}{n_0} \sum_{i=1}^{n_0} Y_{0i}$ among the uninsured, which is necessarily an average of Y_{0i} , with no information on Y_{1i}



Average causal effect (II)

In Table 2 we see the average Y_{1i} only for the insured and the average Y_{0i} only for the uninsured. This corresponds to the **difference in means** by treatment status:

$$Avg_n[Y_i | D_i = 1] - Avg_n[Y_i | D_i = 0] = Avg_n[Y_{1i} | D_i = 1] - Avg_n[Y_{0i} | D_i = 0] \quad (1)$$

- We are after an average causal effect over everyone's Y_{1i} and Y_{0i} ,
- but we only see the average Y_{1i} for the insured and the average Y_{0i} for the uninsured

This leads to a **biased** estimate of the effect of HI on health



Constant-effects assumption

Assume that HI makes everyone healthier by a constant amount τ (the true causal effect, positive):

$$Y_{1i} - Y_{0i} = \tau \iff Y_{1i} = Y_{0i} + \tau$$

Substituting in (1):

$$\begin{aligned} Avg_n[Y_{1i} \mid D_i = 1] - Avg_n[Y_{0i} \mid D_i = 0] &= Avg_n[Y_{0i} + \tau \mid D_i = 1] - Avg_n[Y_{0i} \mid D_i = 0] \\ &= \tau + \{Avg_n[Y_{0i} \mid D_i = 1] - Avg_n[Y_{0i} \mid D_i = 0]\} \end{aligned}$$

The comparison of insured and uninsured equals the causal effect τ plus the difference in average Y_{0i} between insured and uninsured.

Difference in group means = Average causal effect + **Selection bias**



Selection on observables

- When the only source of selection bias is a set of differences in **observable** characteristics, the identification problem is easily fixed
- We can focus on samples of people with the same characteristics
- E.g. if the only source of bias is education, we can restrict the sample to insured and uninsured college graduates only

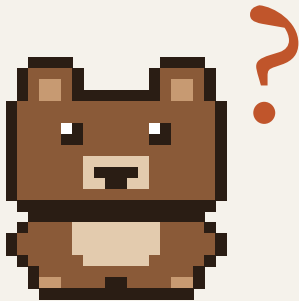


Selection on unobservables

- However, with many observable differences (education, income, age), as in Table 2, it is reasonable to expect **unobserved** differences too
- Even if we hold observed characteristics fixed, the insured and uninsured would differ in ways we do not see

The main problem in programme evaluation is eliminating the selection bias arising from **unobserved differences**





Questions?

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